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**SUBJECT CODE NO: RNP-8019**  
**FACULTY OF SCIENCE AND TECHNOLOGY**  
**S.E (CSE/CS/CE/AI&DS/AIML/IT/AI) (NEP) Sem - III**  
**Examination May/June-2026**  
**Discrete Mathematics**

**[Time: 1:30 Hours]****[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B

- 1) Part A: Question No. 1 is compulsory.
- 2) Attempt any four questions from remaining Part B: Q.2 to Q.7.

**PART-A****Q1 Solve any 5 following questions****10**

- a) Define power set.
- b) What is conditional probability?
- c) Define logical equivalence.
- d) What is recursive definition?
- e) Define partial order relation.
- f) State Lagrange's theorem.
- g) What is parity bit?

**PART-B****Solve any four questions from the following.**

- |                                                                                                                                |          |
|--------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Q. 2</b> Prove De Morgan's Law.                                                                                             | <b>5</b> |
| <b>Q. 3</b> If $P(A) = 0.5, P(B) = 0.3, P(A \cap B) = 0.2$ , find $P(A   B)P(A   B)P(A   B)$ .                                 | <b>5</b> |
| <b>Q. 4</b> check whether $(p \vee q) \rightarrow r$ is logically equivalent to $(p \rightarrow r) \wedge (q \rightarrow r)$ . | <b>5</b> |
| <b>Q. 5</b> Translate: "All students are intelligent" using quantifiers.                                                       | <b>5</b> |
| <b>Q. 6</b> Define one-one and onto function with example.                                                                     | <b>5</b> |
| <b>Q. 7</b> Explain parity check with example.                                                                                 | <b>5</b> |